

Evaluation:

Research:

At the beginning of this project, I carried out research into game mechanics, a series of games, level designs, themes, 3 books of my choice and engagement. I completed a good amount and an excellent quality of research as I explored 3 different books which I found in the library. I then expanded the 3 books that I found by researching 5 games which I then used to further refine and develop my game ideas to get ready to present them to the class.

The research which I conducted showed me that endless runner games are simple in terms of mechanics as the player is already running automatically which limits the variety of mechanics which can be added to the game. This means that I had to spend more time building mechanics, animations, obstacles and enemy ais. I did this because even if my game is limited in one aspect, I can push the other aspects as much as possible to still create an engaging game.

I used this research in my game by applying it and creating outstanding animations that play when the game starts and ends to ensure a smooth start and end to my game. I even created a camera shake for when the player collides into an object to really immerse the player into the game and to create that collision effect. I also added coins into the game to create a reward for the player to chase in the game.

Overall, my research created a good foundation and pillars to help develop my game as smoothly and as safely as possible, making sure that I meet my deadlines. However, I wish I had researched deeper into advanced unity mechanics/code and applied the to my game, but I am still happy with my progress this year in 3d development.

Design/ideation and prototyping:

During the design stage of my project, I created my ideas for my endless runner game called Space Run. My ideas initially came from the research which I conducted, but I then further refined and developed them through my designing and prototyping phase of this

project. I developed ideas for my lava planet space themed game, advanced Ui, obstacles and a good, paced game. The ideas I created from these aspects helped me shape up my game to ensure I knew exactly what I was going to do once in the development stage again.

Overall, I was happy with my designs and believe I produced enough planning and prototyping to support the development stage to the best of my ability. I created concepts for my theme, menus, and gameplay systems which gave me a good overview before developing. I also experimented in my spare time on unity on potential animations I could add to my game which I did end up adding in which I think was the highlight of my prototyping phase as I experimented on unity to find suitable animations and then applied them to my game when the development stage began.

I do believe I did a good amount of detail in this phase of the project; this is proven by my games design document. However, I could have drawn some sketches of the things I was going to add to my game as it would have really completed this phase of the project.

My final project matched most of the designs that I had however they were a few things which I wish I did add from my design phase such as a climbing mechanic to create a whole new area and ideas for the player to explore and take part in. I also wanted to add an advanced shooting mechanic to the player to allow him to actually eliminate the enemy ai before they get the chance to kill the player however due to the limitations in an endless runner and the mechanics of my enemy ai I did not know how to add this to my game. I do believe if I had more time, I could have added these into the game, especially the climbing mechanic, but I did the best that I could with the time given to me.

Final product:

I am satisfied with how my game turned out as I successfully created an endless runner game called space runner that includes core gameplay, collectibles, menus and design from my research and design phases. Throughout the project I did apply my research and design that I did into my game which I am happy about as in the past projects I would do some research and then develop and end up doing whatever I could instead of following a proper made plan and blueprint.

The features of my game, which I am most proud of, are my animations for the past 2 years. This has been something which not only struggled to implement, but I also struggled to understand how they worked on unity. I am an enthusiastic fan of the Collision animation, the start animation and the death animations that play in the game as they create a good user experience.

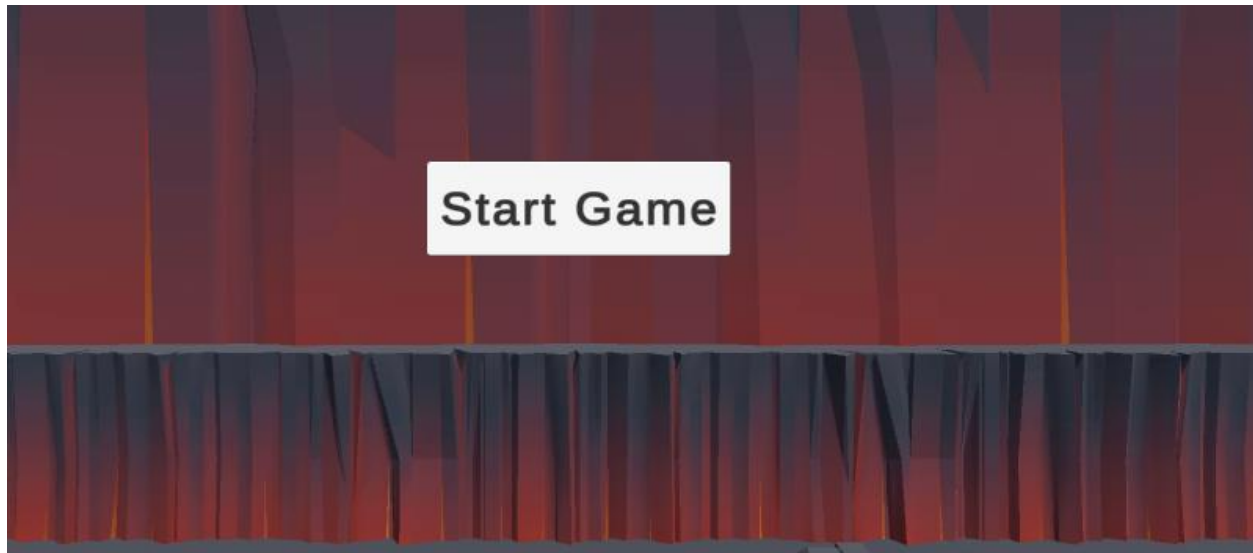
However, there are some areas of the final product which could have been up to a higher standard and or to the same standard as other aspects of the game for example the game has 3 segments which infinitely generate randomly. INITIALLY THE USER WON'T REALISE THAT THEY ARE PLAYING THE SAME 3 SEGMENTS AGAIN AND AGAIN JUST IN DIFFERENT FORMATS HOWEVER AFTER A FEW MINUTES OF RUNNING THE GAME IT BECOMES EASY TO INTERPRET AND CAN MAKE THE GAME BORING AND REPETITIVE.

If I had more time, I would have created more segments in the game, even with different obstacles and mechanics on the enemy ai's. This was the weakest part of my game however I am still confident that it is a good enough standard for me to get the game to a good grade, but in terms of my own standard I wish I created more segments earlier on.

The testing which I conducted throughout the development of my game which is proven I'm my development log allowed me to take a step back and to analyze how my game was looking from a different point of view and it allowed me to change aspects in my game like the jumping which was too weak. Because my game is set in space, it made sense to create a low-gravity jump which allowed the player to jump higher and slowly come back down.

If I was given more time to complete this project or could do this project again I would have a deeper planning in my development stage and aim to have all aspects of the game at a consistent quality and ensure that I have weekly goals set up on my Trello before I begin the development to ensure I stay on track and even if I was to go off track from my development plan I would just work harder and longer the following week to get back on track and complete the goals I set myself.





Skills Development:

Throughout this project I learnt a lot of new skills not only in the development stage but also in the research, designing, and planning phases of my entire project. My understanding of the process of creating a game is a lot better now as I know the full process that takes place from the very start to the end. This project also improved my confidence in unity as the more hours I spend on unity and add to my experience the more confident and the sharper my skills will get.

The skill that I picked up which I am the proudest about is my fluency in animation creation and most importantly understanding. I really struggled in past projects to understand how animations worked but now after hours of working and reining my skills I have managed to learn about them and implement them to a good standard. This is proven in my game as I have advanced animations in my menu, in the players death and when the player collides into something.

I put a good amount of time into my research and design phases as I conducted a lot of research which helped me a lot when it came to understating the game which I wanted to make and what I am that the game will include. When I completed my games design document, I was confident about what my game was and what it would be about, which

allowed me to keep my head up throughout the development stages without feeling nervous about the approaching deadline. I am also proud of the fact that in my spare time I was experimenting on unity with mechanics and game systems to see what works and what did not work as it allowed me to improve my research and my game.

As I look ahead in my games development journey in the university of Huddersfield a hopefully a Carrer one day I would really like to know more about advanced mechanics and better level designs and game systems. Something which I think is more crucial than implementing these into games is understanding them because I can apply them to different circumstances and I will know that my code and design are doing in the game.

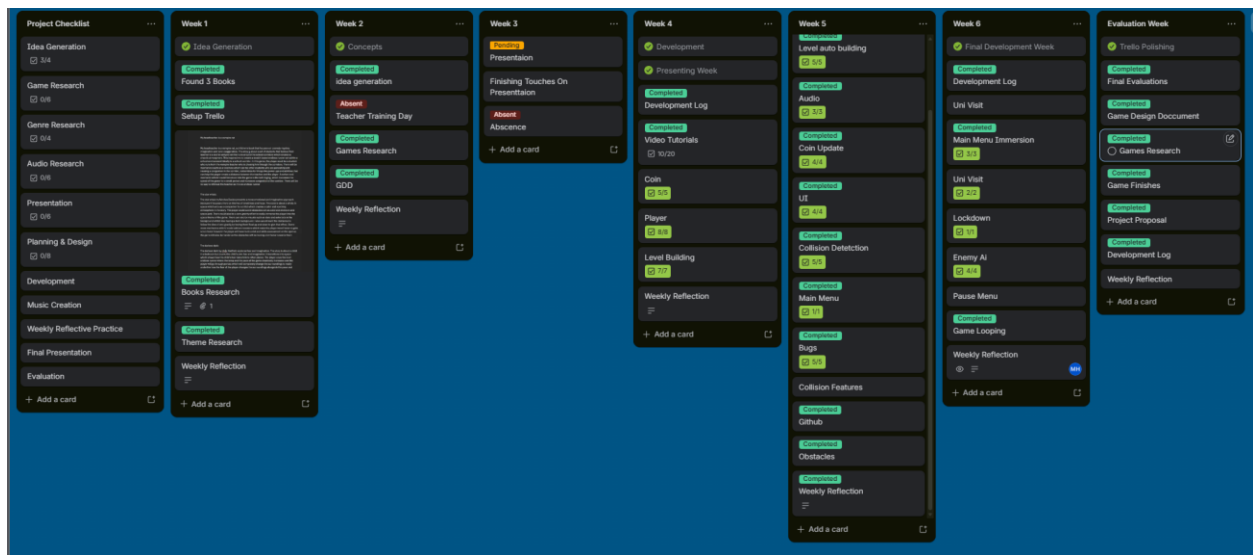
Project Reflection:

This project went well as I developed a 3D endless runner that followed the planning and design that I had created. I also developed my skills further in this project hand I did with the past projects which is due to the pressure of the final major project and the fact that I know that university I am approaching I can imagine the workload and the difficulty increases drastically.

I managed my time exceptionally well throughout the project consistently by using my Trello board and by watching the deadlines for each stage of the project and meeting them accordingly .However there were aspects in my development process which took longer than others such as creating an enemy ai this could be due to my skills and experience which is expected as I only have 1 year of unity 3D experience, this taught me how to prioritize certain task to optimize my development stage.

If I could go back, I would spend more time experimenting on unity as I feel like this saved me a lor of time I the development stage as I had lots of spare time in my research and design phases of the project. I would also implement this in other project I did if I had the hence, but I will most definitely be doing this in future project sin university and Carrers.

Overall, this project and my academic years in this college have really helped and supported me in my b=start into this career by improving my understanding of the full process of a games project. I have gained valuable experience in unity, research, designing and prototyping, documenting and project management and organization skills using applications like Trello. I believe all of this will benefit me in my education journey in university and my Carrer as a game's designer in the future.



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